Refraction

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect a change in wave speed to wavelength and direction.

SUCCESS CRITERIA - I can...

1. Measure ray angles from the normal.
2. Explain frequency continuity at a stationary boundary.
3. Predict bending toward or away from the normal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “refraction” and “normal line” to explain your model or result. Sentence starters are optional.

refraction: change in wave direction when speed changes at a boundary

normal line: a reference line perpendicular to a surface

medium: the material through which a mechanical wave travels

01 Warm-up: bring back an earlier idea

A wave has period 3 s.
Find the cycles completed in 15 s and its frequency.

Word help: frequency = number of cycles each second

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

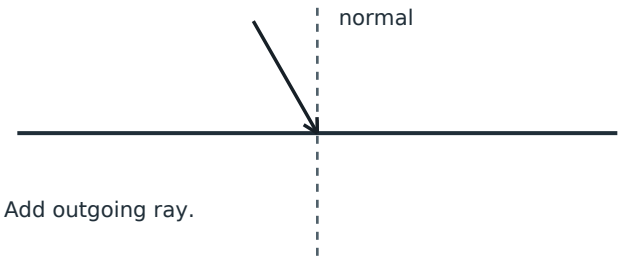
02 Worked example: explain the model

Reference: $v_1 = f\,\lambda_1$; $v_2 = f\,\lambda_2$; $\lambda_2/\lambda_1 = v_2/v_1$.
Stationary boundary: frequency stays constant. A slower medium bends an oblique ray toward the normal.

Worked reasoning: Light enters a slower medium at an angle to the normal and bends toward the normal. The normal is perpendicular to the boundary.

Your turn: Draw and label the boundary, normal, incident ray and refracted ray.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add outgoing ray.

Marzano check - circle: ① ② ③

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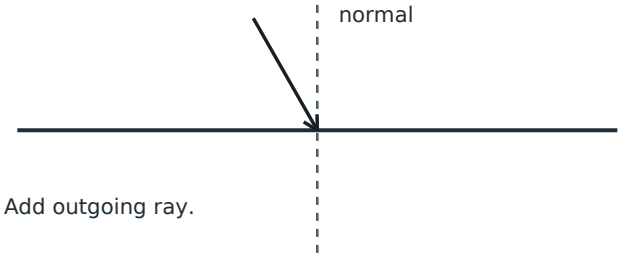
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Mark the normal at the boundary. A ray enters a slower medium at an angle.
Sketch the refracted direction.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add outgoing ray.

Marzano check - circle: 1 2 3

04 Apply the physics

A wave crosses into a slower medium. Its speed changes from 12 to 6 m/s while frequency stays 3 Hz.
Find the wavelength on each side.
Draw wavefronts with the correct relative spacing.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add wavefronts, direction and wavelengths.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A student measures an incidence angle from the surface instead of the normal. If surface angle is 20°, what angle belongs in Snell’s law?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

Samples at 14°C and 56°C are mixed with no energy input or phase change. A student reports 68°C at equilibrium. Is that plausible?
Explain.
Label temperatures, samples and relevant energy transfers.

Sentence starter: My claim is … . The diagram/data show …, so … .

Sample A

Sample B

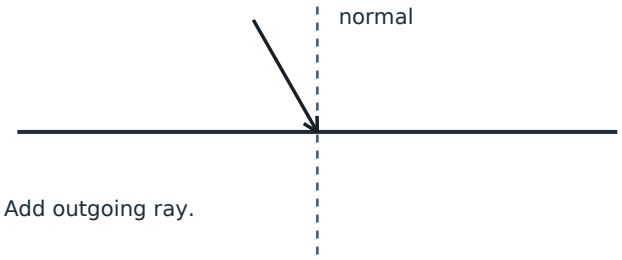
Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

07 Exit check

A ray enters a faster medium at an angle. Does it bend toward or away from normal?

Sentence starter: I notice … . This supports … because … .



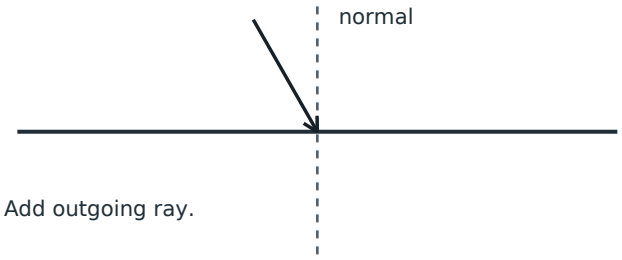
Add outgoing ray.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Draw two rays entering a slower medium: one obliquely and one along the normal.
Explain why one changes direction and the other does not.

Sentence starter: My claim is … . The diagram/data show …, so … .



Add outgoing ray.

Marzano check - circle: 1 2 3